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Transit Light Curves, Exoplanet Detection, and Atmospheric Studies

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ABSTRACT

HD 209458b is a well-studied hot Jupiter and a key target for both exoplanet detection and atmospheric characterization. In this work, we first applied machine learning models: Support Vector Machines (SVM) and Artificial Neural Networks (ANN) to detect planetary transits in Kepler light curves. The models performed well on synthetic data but struggled with real observations, highlighting the challenges of noise and variability. We then modelled and retrieved the planet's atmospheric properties using TauREx 3. Retrievals on both synthetic and observed spectra revealed strong constraints on temperature, radius, and molecular abundances, particularly H_2O and CO_2. Our results shows how machine learning and atmospheric modelling can work together to improve our understanding of exoplanets.

Key words. exoplanets – transit light curves – machine learning – TauREx

1. Introduction

HD 209458b is one of the most important exoplanets, as it was first detected using the radial velocity method and later confirmed through the transit method Charbonneau et al. (1999), making it the first exoplanet to be recognized as a confirmed transiting planet around its host star. This planet is a hot Jupiter which orbits very close to the main-sequence, solar-type star HD 209458 having a spectral type of G0V, at a semi-major axis of 0.046 AU. It has a very short orbital period of 3.5 days, and its mass has been measured to be 0.62 M_{Jup}. Using transit photometry, a $1.58 \pm 0.18\%$ dimming of the starlight was observed, from which the planet's radius was calculated to be $1.42 \pm 0.10\, R_{Jup}$ Henry et al. (1999).

Hot Jupiters are very interesting and unique due to their peculiar atmospheric features. Many of them have exceptionally large radii which is probably related to their atmospheres. Since they are much closer to their host star, the intense radiation from the star causes the atmosphere of the planet to expand. Also, their atmospheres have a mix of quite complex molecules such as ions, atoms, molecules, and even cloud particles (condensates). In addition, such planets have large temperature differences between their day and night sides and there is a possibility that the velocity of the super-fast winds may exceed the speed of the sound Fortney et al. (2021).

A previous study on HD 209458b revealed that the atmospheric conditions on the planet's day-side are unique, as the temperature increases with altitude, leading to a thermal inversion and additionally, a stratosphere has been detected. These two findings suggest an unusual atmospheric composition, as the inversion is caused by an unknown absorber at high altitudes that captures light from the star. Furthermore, the extreme heat in the upper atmosphere causes the behavior of water vapor to emit light instead of absorbing it, reversing its typical behavior Burrows et al. (2007). In addition to this, another prior research on this planet revealed that hydrogen escaped from its upper atmosphere which explains the mass loss due to the intense radiation of the star Bourrier & Des Etangs (2013).

HD 209458b remains a standard for testing atmospheric models and retrieval methods due to its diverse atmospheric behaviour, notable transit features, and well-characterized system parameters. Its strong transit signal, close-in orbit, and complex atmospheric behaviour make it an ideal target for detailed modelling and retrieval analysis. In this study, we use HD 209458b as a case study to simulate and retrieve atmospheric properties through both synthetic and real spectra. By applying tools like TauREx, we aim to explore its molecular composition, thermal structure, and how well different modelling techniques can reproduce its observed atmospheric behaviour. Additionally, we investigate the performance of machine learning methods, specifically Support Vector Machines and Artificial Neural Networks, in detecting planetary transits within light curve data.

2. Stellar and Planetary Parameters of HD 209458b

The system properties of HD 209458b, a transiting hot Jupiter orbiting a solar-type star, make it an ideal candidate for transit photometry and atmospheric characterization. The host star is a G0V main-sequence star with an effective temperature of 6050 K, which provides excellent photometric stability and brightness for high-

precision measurements Stassun et al. (2017). The planet orbits very close to the star, at a distance of 0.047 AU, completing an orbit every 3.52 days Charbonneau et al. (1999). It has a very large radius of 1.39 R_{Jup} which results in a significant transit depth of about 1.64%, allowing for accurate detection of both transit and secondary eclipse signals Brown et al. (2001). Its short orbital period allows for the observation of prominent and frequent transits within a short amount of time, which is useful for both transmission and emission spectroscopy.

The stellar and planetary parameters used in this study are summarized in table 1 and table 2 respectively. These values were primarily obtained from the NASA Exoplanet Archive, as provided by Stassun et al. (2017). Observations from HST and Spitzer have also revealed key atmospheric constituents in HD 209458b, including H_2O, CO, and possibly CO_2 Swain et al. (2009). The dayside atmospheric temperature is estimated to be approximately 1200 K, based on infrared emission measurements from Spitzer Deming et al. (2005). These system and atmospheric properties are widely used in modeling studies to simulate the spectrum of HD 209458b and to better understand its thermal structure and energy distribution.

Table 1. Stellar parameters for the host star HD 209458.

Parameter	Value
Stellar effective temperature $T_{\rm eff}$	6050 K
Spectral type	G0V

Table 2. Planetary parameters for the exoplanet HD 209458b.

Parameter	Value
Planetary radius $R_{\rm p}$	$1.39\,R_{\rm Jup}$
Semi-major axis a	0.047 AU
Planetary mass $M_{\rm p}$	$0.73\,M_{\rm Jup}$
Orbital period P	3.52 days
Atmospheric composition	H_2O, CH_4,
CO_2, CO (randomized abundances)	
Temperature T	1200 K

Using the parameters presented in table 2, we generated a transit light-curve of our exoplanet with `batman` Kreidberg (2015) (shown in figure 1).

3. Detection of Transit Light Curves Using Machine Learning

In the study of exoplanets, one of the biggest challenges is managing the huge volume of photometric data collected by space missions like Kepler. This makes it difficult and impractical to manually inspect light curves for signs of planetary transits. When a planet passes in front of its host star, it causes a small periodic dip in the star's brightness, which appears in the light curve. However, this signal can easily be hidden by stellar activity or instrumental noise. To handle this complexity, we use machine learning, which has become an essential tool for exoplanet detection. Supervised models such as Support Vector Machines (SVMs) and Artificial Neural Networks (ANNs) are especially helpful because they can be trained to recognize the shapes of transits in light curves, making the detection process faster and more accurate.

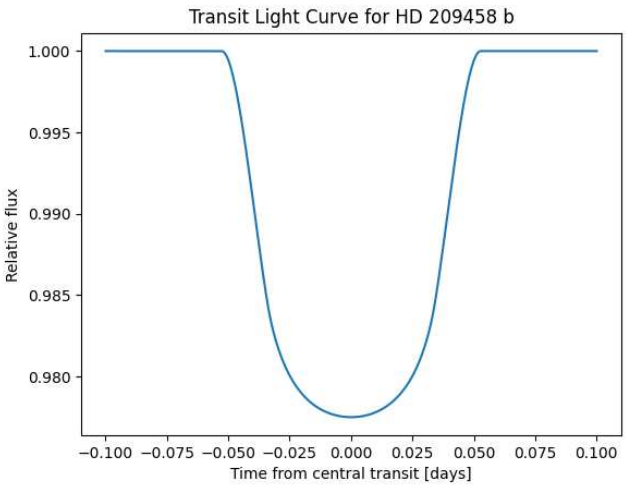

Fig. 1. A `batman` generated transit lightcurve of HD 209458 b.

3.1. Datasets

For this study, we used two datasets derived from the Kepler mission which detected the transit of exoplanets by observing the stellar flux over time. The datasets used are:

- **Kepler data with injection** (`kepler/data_injection`): This dataset contains light curves where synthetic planetary transits were injected into real Kepler data to simulate various scenarios. It is mainly used to train models and check how well they can detect planets.
- **Kepler data without injection** (`kepler/data_no_injection`): This dataset consists of light curves containing real photometric observations that include both known exoplanetary transits and curves dominated by stellar variability or noise. No artificial transits are injected, and the dataset is used for validation and testing.

Each dataset is divided into two parts: training and testing. The light curves are labeled using a binary classification: label 2 indicates a system with a planet (i.e., a transit is present), and label 1 indicates a system without a planet (no transit). In the injection dataset (`kepler/data_injection`), the training set contains 2861 systems with a planet and 2,226 without, while the test set includes 316 with a planet and 254 without. In total, this dataset includes 5,657 light curves. In the non-injection dataset (`kepler/data_no_injection`), the training set consists of 37 systems with a planet and 5050 without, while the test set includes 5 with a planet and 565 without. This gives a total of 5,657 light curves as well. The non-injected dataset contains more examples of systems without planets than with planets.

Each light curve contains around 3,197 brightness values measured over time. These are normalized and centered around the expected transit. The goal is to classify each

light curve as either showing a planetary transit (positive) or not (negative).

3.2. Methodology

Machine learning (ML) is particularly suitable for detecting planetary transits in large photometric datasets, such as those produced by the Kepler mission. Traditional transit detection algorithms rely on manual inspection or simple thresholding methods, which can become inefficient and error-prone when applied to millions of light curves.
In contrast, ML techniques can automatically learn complex patterns associated with transit signals, even when those signals are weak or partially obscured by stellar noise. This is especially valuable in addressing the following challenges:

- **Stellar variability**: Variations in stellar brightness due to starspots, flares, or pulsations can mimic or obscure transit signals.
- **Instrumental noise**: Systematic errors introduced by the telescope or detector may produce spurious signals or false positives.
- **False positives**: Phenomena such as eclipsing binaries or background contamination can resemble planetary transit signals.

By training on labeled datasets, machine learning models can learn to distinguish between real transits and these various sources of noise or false positives. In this study, we used two supervised machine learning techniques such as Support Vector Machines (SVM) and Artificial Neural Networks (ANN) to classify the light curves. These models are able to capture subtle patterns that help distinguish planetary events from other similar signals. This improves detection accuracy and reduces the amount of manual workload required for each potential planet.

3.2.1. Support Vector Machine (SVM)

Support Vector Machines (SVMs) are supervised learning algorithms effective for binary classification problems such as distinguishing planetary transits from non-transits. In this study, we applied SVMs with three different kernels: Linear, Gaussian (Radial Basis Function, RBF), and Polynomial (degree 4).
Before training the SVM, we preprocessed the light curves using a custom pipeline. We first, transformed each light curve into frequency domain using Fourier Transform to highlight the periodic patterns. Then, we applied normalization to scale the data, followed by Gaussian smoothing to reduce noise and make the transit features more prominent. Finally, the features were standardized using `StandardScaler` so that each feature has a zero mean and unit variance, and this ensures consistency in the input for the model. These steps help to prepare the data and improve the performance for effective classification.

The Support Vector Machine works by finding a boundary or hyperplane that best separates the two classes while maximizing the margin which is the distance between this boundary and the closest data points from each class, called support vectors. This concept can be expressed as optimization problem given by,

$$min_{w,b} = \frac{1}{2}||w||^2$$

subject to,

$$t_n(w^T\phi(x_n) + b) \geq 1$$

where, w is the is the weight vector, b is the bias, $\phi(x_n)$ is the input and $t_n \in -1, +1$ is the class label. This ensures that all data points are correctly classified and lie at least a distance of 1 from the decision boundary, while minimizing the complexity of the model.

The SVM models were trained and tested separately on both the injected and non-injected datasets. This helped us observe and evaluate how well the models performed in detecting clear, synthetic transit signals, as well as more realistic, noisy light curves from the original Kepler observations. For each of the data sets, we use training and testing data, where each row in the dataset is a light-curve. We separated the data into input features and labels where the labels indicate whether the light curve shows a planetary transit (label 2) or not (label 1). To better understand the light curve structure, we visualized a randomly selected sample labelled as a planet. This helped us confirm the expected light curve shape and check for obvious data quality issues.

Before training, we applied the preprocessing pipeline to both the training and validation datasets as mentioned previously. This resulted in the reorganization of their features and labels. The labels were also converted into a simplified binary format where `True` represents a planetary transit and `False` represents a non-transit.

To evaluate how different SVM models perform in detecting planetary transits, we trained three separate models using different types of decision boundaries called kernels, as previously mentioned:

- **Linear kernel**: Simple linear separation of the data.
- **Gaussian (RBF) kernel**: Non-linear decision boundary, with gamma set to ‘scale‘ (default behaviour).
- **Polynomial kernel (degree 4)**: Higher-dimensional mapping with degree 4, allowing more complex separation.

Each of the three models was trained on the injected dataset and tested on it. After training, predictions were made for both the training data and the validation data to see how well each model learned and how well it generalized to the unknown data.
We used several metrics to evaluate the performance of the model:

- **Accuracy:** Tells us how many total predictions were correct.
- **Precision:** Tells us how many predicted planets were actually correct.
- **Confusion matrix:** A table showing true negatives, false positives, false negatives, and true positives.

These metrics allowed us to understand how well each kernel performed at identifying planetary transits and how often it misclassified non-transits either by missing real

planets or falsely identifying noise as a planet. The results for each model are provided in section 3.3. For each kernel, we recorded its precision scores and confusion matrices for both training and validation sets. This allowed us to compare all three models side by side and identify which one was best at detecting planetary signals while avoiding false positives. We then repeated the same process using the dataset with non-injected planetary transits.

3.2.2. Artificial Neural Network (ANN)

Artificial Neural Networks (ANNs) are machine learning models based on the linked structure of neurons in the human brain. They are especially useful for classification tasks requiring complicated patterns to be learnt directly from data. In this study, we rely on a feedforward neural network, also known as a multilayer perceptron (MLP) to classify planetary transits in light-curve data. The ANN model uses the same preprocessing pipeline that was applied for the Support Vector Machine (SVM) models (see Section 3.2.1).

To reduce computational time and allow for faster experimentation, we selected a subset of 1000 light curves from the training set and 200 from the validation set. These subsets were chosen to keep the training process manageable while still maintaining a reasonable balance between planet and non-planet classes. To further address residual class imbalance, we used the Synthetic Minority Over-sampling Technique (SMOTE), which generates new synthetic samples of the minority class by interpolating between existing samples. This enhancement helps the model to better learn patterns associated with planetary transits without overfitting to duplicated data.

The ANN architecture consists of:

- An input layer matching the number of features in the processed lightcurve data.
- Two hidden layers with 64 and 32 neurons. While a single hidden layer is theoretically sufficient to approximate any function, deeper networks are often more efficient and generalizable. The first layer detects broad patterns in the data, while the second layer refines these patterns to support the ultimate prediction. Both hidden layers use the ReLU (Rectified Linear Unit) activation function which introduces non-linearity and improves learning performance. The activation is defined as::

$$z_j = \mathrm{ReLU}\left(\sum_i w_{ji}^{(1)} x_i + w_{j0}^{(1)}\right)$$

  ReLU enables the network to model complicated interactions and helps reduce the the vanishing gradient problem, which can occur in deep networks. It also encourages sparse activations, which improves the model's efficiency.
- A Dropout layer with a dropout rate of 0.2 is applied after each hidden layer. This helps reduce the risk of overfitting by randomly deactivating 20% neurons during each training step.
- An output layer consisting of a single neuron with a sigmoid activation:

$$y_k = \sigma\left(\sum_j w_{kj}^{(2)} z_j + w_{k0}^{(2)}\right)$$

  where $\sigma(a) = \frac{1}{1+e^{-a}}$. Here, $k = 1, ..., K$ and K is the total number of outputs. The sigmoid activation function at the output layer maps the network's output to the interval [0,1]. This ensures that the result can be interpreted as the probability that a transit is present in the input light curve. To convert this probabilistic output into a binary prediction, a threshold of 0.5 is applied:
  - if the output is greater than 0.5, the sample is classified as a transit event (label 2);
  - otherwise, it is classified as a non-transit (label 1)

The network was trained using the Adam optimizer, which is an adaptive gradient-based method well-suited for sparse and noisy data. The binary cross-entropy loss function was used, as it is appropriate for binary classification tasks such as distinguishing between transit and non-transit light curves. Training was run for up to 30 epochs; however, to avoid over-fitting, early stopping was applied based on validation loss, halting training if performance stopped improving.

The performance of the ANN model, including its precision on both the injected and no-injection datasets, is summarized in the results section (Section 3.3).

3.3. Results

The performance of the Support Vector Machine (SVM) classifiers and the Artificial Neural Network (ANN) model was evaluated through two primary evaluation metrics: the confusion matrix and the precision score. The confusion matrix provides a detailed breakdown of true negatives, false positives, false negatives, and true positives, allowing us to understand the types of errors made by the model. Precision, which is defined as the ratio of true positives to the sum of true positives and false positives, was chosen as the main quantitative metric because, in exoplanet detection, minimizing false positive identifications is crucial to avoid misclassifying stellar variability or noise as planetary transits. Both metrics were computed separately on the training set and the development (validation) set to evaluate the model's learning ability and generalization performance.

As shown in Table 3, the Support Vector Machine (SVM) models achieved low precision values around 55–56% on the injected dataset, but failed to generalize to the non-injected (real) dataset, where all kernels dropped to nearly 0% precision on the validation set. This suggests that while the models could detect synthetic transit signals, they overfitted the training data and struggled with the noise and variability present in real Kepler light curves.

The Artificial Neural Network (ANN) outperformed the SVMs on the injected dataset, reaching 70% precision on the training set and 21% on the validation set (see Table 4). However, it too failed to detect real transits on the non-injected dataset, again indicating poor generalization. Despite this, the ANN showed a slight advantage over

Table 3. Confusion matrices and precision scores for SVM models with different kernels on injected and non-injected datasets.

Kernel	Linear	RBF	Poly
	No injection data		
Train	$\begin{bmatrix} 5050 & 0 \\ 0 & 37 \end{bmatrix}$	$\begin{bmatrix} 5050 & 0 \\ 1 & 36 \end{bmatrix}$	$\begin{bmatrix} 5050 & 0 \\ 0 & 37 \end{bmatrix}$
Precision	1.0	1.0	1.0
Dev	$\begin{bmatrix} 565 & 0 \\ 5 & 0 \end{bmatrix}$	$\begin{bmatrix} 565 & 0 \\ 5 & 0 \end{bmatrix}$	$\begin{bmatrix} 559 & 6 \\ 5 & 0 \end{bmatrix}$
Precision	0.0	0.0	0.0
	Injected data		
Train	$\begin{bmatrix} 0 & 2226 \\ 0 & 2861 \end{bmatrix}$	$\begin{bmatrix} 37 & 2189 \\ 14 & 2847 \end{bmatrix}$	$\begin{bmatrix} 95 & 2131 \\ 50 & 2811 \end{bmatrix}$
Precision	0.56	0.56	0.57
Dev	$\begin{bmatrix} 0 & 254 \\ 0 & 316 \end{bmatrix}$	$\begin{bmatrix} 0 & 254 \\ 0 & 316 \end{bmatrix}$	$\begin{bmatrix} 55 & 199 \\ 63 & 253 \end{bmatrix}$
Precision	0.55	0.55	0.56

SVMs, likely due to its ability to capture more complex patterns through its multi-layer architecture and dropout regularization.

For the SVM model, the accuracy on the training set and testing set of the injected dataset is quite similar for the three kernels with an average of 55-56% and among the three the polynomial kernel has the best accuracy. For the non-injected dataset, the accuracy on both the training and test set for the three kernels are quite close with an average of 98-99% which mostly indicates over-fitting or class imbalance since precision is 0.
For the ANN model, the accuracy on the training set of the injected dataset is $\sim 78\%$ while for the testing set it is $\sim 31\%$. For the training set of the non-injected dataset, the accuracy for both the training and test set is $\sim 99\%$ but for the test set it clearly indicates over-fitting or class imbalance due to 0 precision.

There are several ways this work could be improved in the future. The models could be made more advanced, for example, by using deeper neural networks or adding convolutional layers that are better at capturing patterns in the data. The preprocessing steps could also be improved by adding better noise removal or filtering techniques to clean up the light curves. Using larger and more balanced datasets would help the models learn more general patterns and avoid over-fitting. It could also help to try data augmentation, which means creating slightly modified versions of the data to make the model more robust.

Overall, both models performed significantly better on the synthetic (injected) data, highlighting the challenges of applying machine learning to real astronomical signals, which often have lower signal-to-noise ratios and more variability. While SVM and ANN showed promise in detecting clear, artificial transits, more advanced techniques and higher-quality datasets will be necessary to achieve reliable transit detection in real observational data.

Table 4. Confusion matrices and precision scores for ANN on injected and non-injected datasets.

	No injection dataset
Train	$\begin{bmatrix} 970 & 19 \\ 0 & 989 \end{bmatrix}$
Precision	0.98
Dev	$\begin{bmatrix} 198 & 0 \\ 2 & 0 \end{bmatrix}$
Precision	0.0
	Injected dataset
Train	$\begin{bmatrix} 524 & 365 \\ 27 & 862 \end{bmatrix}$
Precision	0.70
Dev	$\begin{bmatrix} 27 & 133 \\ 5 & 35 \end{bmatrix}$
Precision	0.21

4. Atmospheric Study on Planet

4.1. Model

To simulate the atmospheric characteristics of HD 209458b, we used the TauREx 3 (Al-Refaie et al. (2021) & Al-Refaie et al. (2022)) forward modelling framework to generate a synthetic transmission spectrum. We obtained the planetary and stellar parameters from the NASA Exoplanet Archive. We created a TauRex `.par` file to model a simple forward transmission spectrum. Within this model, we assumed that the atmosphere is composed primarily of hydrogen and helium with a ratio of 4:1, consistent with the expected composition of gas giants. In addition to this, four trace gases H_2O, CH_4, CO_2 and CO were included and their abundances were randomly selected within a range of 10^{-8} to 10^{-2} which are shown in table 5.

Table 5. Gas species and their mixing ratios as defined in the `.par` input file for TauREx. The gas type is set to `constant`, indicating a uniform volume mixing ratio throughout the atmosphere.

Gas	Mixing Ratio
H_2O	1.1185×10^{-4}
CH_4	1.0×10^{-5}
CO_2	5.0×10^{-6}
CO	2.0×10^{-5}

The atmospheric temperature structure was assumed to be isothermal, with a uniform temperature of 1200 K across all the layers. A simple pressure profile was adopted, ranging from 10^{-4} to 10^{6} bar divided into 30 layers. Planetary parameters were fixed using known values (from table 2), with a mass of 0.73 M_{Jup} , a radius of

1.39 R_{Jup}, and an orbital semi-major axis of 0.047 AU. The opacities for molecular absorption were obtained from TauRex-compatible precomputed cross-section files, and Collision Induced Absorption (CIA) from $H_2 - H_2$ and $H_2 - He$ interactions was also included. In addition, we included Rayleigh scattering from H_2 and He in the forward model using TauREx's default behaviour, which applies scattering based on the specified fill gases. We did not apply any cloud model and assumed a clear atmosphere for simplicity in this forward simulation.

The output of the forward model was a high-resolution synthetic transmission spectrum of HD 209458b, representing the squared planet-to-star radius ratio $(R_p/R_s)^2$ as a function of wavelength. This spectrum was plotted as shown in the Figure 2 and stored in the required format, serving as the input for subsequent retrieval experiments.
In addition to the synthetic spectrum, we examined the observed transmission spectrum of HD 209458b, retrieved from the NASA Exoplanet Archive's Firefly interface. The spectrum, which provides the squared planet-to-star radius ratio $(R_p/R_s)^2$ as a function of wavelength along with associated uncertainties, is shown in Figure 3.

The synthetic spectrum, generated at high spectral resolution and assuming a clear atmosphere, displays pronounced absorption features from molecules such as H_2O, CH_4, CO_2, and CO, as well as a Rayleigh scattering slope at shorter wavelengths. In contrast, the observed spectrum exhibits lower resolution.
Notably, the molecular absorption features in the observed spectrum appear shallower and more muted, likely due to the presence of atmospheric clouds or hazes not accounted for in the forward model. Additionally, the wavelength coverage of the observed data is narrower, limiting the visibility of the full range of spectral features predicted by the model. Despite these differences, both spectra exhibit comparable overall trends in the 2.5–5 μm region. This shows how the derived synthetic spectrum is closely accurate to the observed real spectrum.

4.2. Retrieval

4.2.1. Retrieval on Synthetic Spectrum

To validate our retrieval setup, we performed atmospheric retrieval on the synthetic transmission spectrum generated in Section 4.1. The retrieval was carried out using the TauREx 3 framework with the `Nestle` nested sampling algorithm and 50 live points. We retrieved three key parameters: the atmospheric temperature, the planetary radius, and the volume mixing ratio of H_2O. The remaining molecules included in the forward model (CH_4, CO_2, CO) were held fixed, and the number of atmospheric layers was increased to 100 to reduce dimensionality and computational cost.
To guide the retrieval, we defined uniform priors for each parameter in a `.par` file. The temperature was allowed to vary between 300 and 2000 K, the planetary radius between 0.3 and 2 R_{Jup}, and the H_2O abundance between 10^{-12} and 10^{-1}. These prior bounds reflect both physical plausibility and the need to limit the parameter space to values relevant for hot Jupiters like HD 209458b.

The results demonstrated excellent agreement with the known input parameters. The retrieved temperature was $T = 1200.00^{+0.01}_{-0.01}$ K, precisely matching the initial isothermal profile. The planetary radius was retrieved as $R_p = 1.39^{+0.00}_{-0.00}$ R_{Jup}, in close alignment with the input value of 1.39 R_{Jup}. Water vapour was robustly detected with a retrieved abundance of $\log_{10}(H_2O) = -3.97 \pm 0.01$, consistent with the log of the injected value (1.1185×10^{-4}). The posterior distributions for all three parameters were sharply peaked, as illustrated in Figure 4, confirming the absence of significant degeneracies or noise-related broadening. The high accuracy of the retrieval under idealized, noise-free conditions confirms that our pipeline is functioning correctly and that the parameter bounds were well-defined.

4.2.2. Retrieval on Real Spectrum

Following the validation on synthetic data, we applied our retrieval pipeline to the observed transmission spectrum of HD 209458b. The retrieval was carried out using TauREx 3 in combination with the `MultiNest` nested sampling algorithm, using 1000 live points for robust exploration of the parameter space. The retrieval focused on four key parameters: atmospheric temperature, planetary radius, and the volume mixing ratios of H_2O and CO_2. The remaining gases (CH_4, HCN, and C_2H_2) were fixed at low abundances.
The prior bounds were informed by both literature values (shown in table 6) and physical plausibility. Specifically, the temperature was allowed to vary between 300 and 2000 K, the planetary radius between 0.67 and 2.7 R_{Jup}, and the water vapor abundance between 10^{-12} and 10^{-1}. For CO_2, the bounds were tightened to 10^{-6}–10^{-3} to reflect its expected trace in hot Jupiter atmospheres.

Table 6. Adopted system parameters for HD 209458b based on Xue et al. (2024). These values were used to guide the retrieval setup and prior bounds.

Parameter	**Value**
Stellar Parameters	
Stellar Temperature ($T_\star$)	6050 K
Planetary Parameters	
Planetary Radius (R_p)	$1.35^{+0.006}_{-0.007}$ R_{Jup}
Planetary Mass (M_p)	0.69 M_{Jup}
Semi-major Axis (a)	~0.047 AU
Orbital Period	3.52474859 days
Atmospheric Temperature (T)	1088^{+103}_{-88} K
Atmospheric Composition	
Primary Molecules	H_2O, CO_2, CO, CH_4

The retrieval yielded a best-fitting atmospheric temperature of $T = 704.88^{+46.20}_{-36.36}$ K and a planetary radius of $R_p = 1.38$ R_{Jup}, consistent with the literature. The log of the retrieved H_2O abundance was $\log_{10}(H_2O) = -1.37^{+0.07}_{-0.08}$, and for CO_2, $\log_{10}(CO_2) = -5.25^{+0.23}_{-0.24}$. These values suggest a water-rich atmosphere and a detectable level of CO_2, supporting the oxygen-rich, super-solar metallicity scenario previously reported.

The posterior distributions, shown in Figure 5, reveal broader uncertainties than those from the synthetic

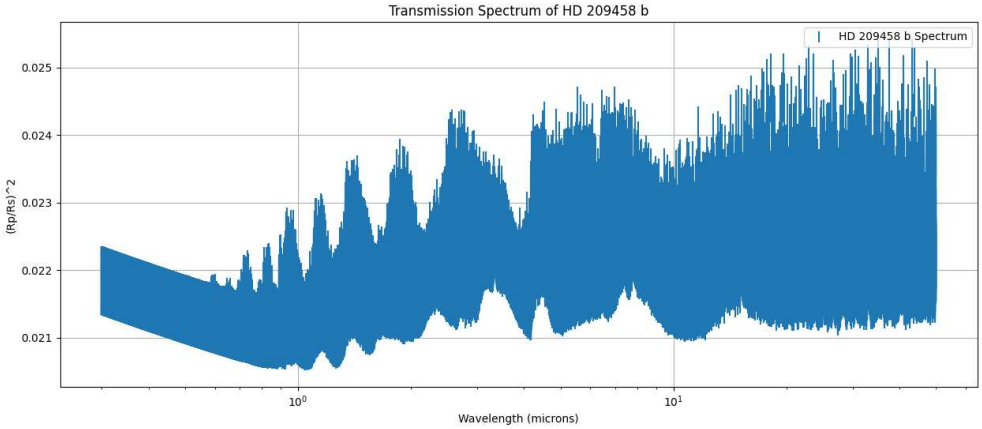

Fig. 2. Synthetic transmission spectrum of HD 209458b generated using the TauREx forward model

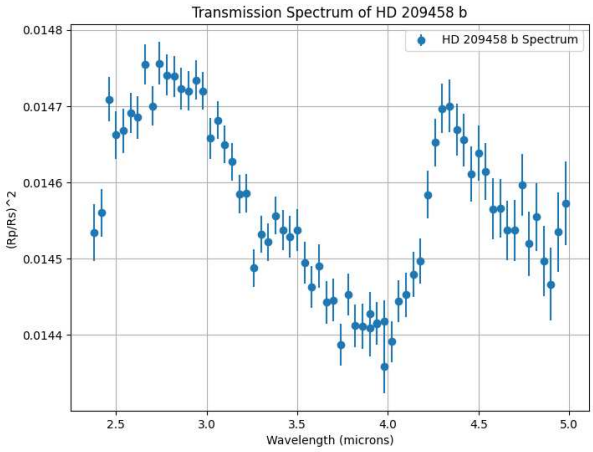

Fig. 3. Observed transmission spectrum of HD 209458b retrieved from the NASA Exoplanet Archive.

retrieval, reflecting the lower resolution and higher noise level of the observed spectrum. Nevertheless, the retrieval provides valuable constraints on key atmospheric components.

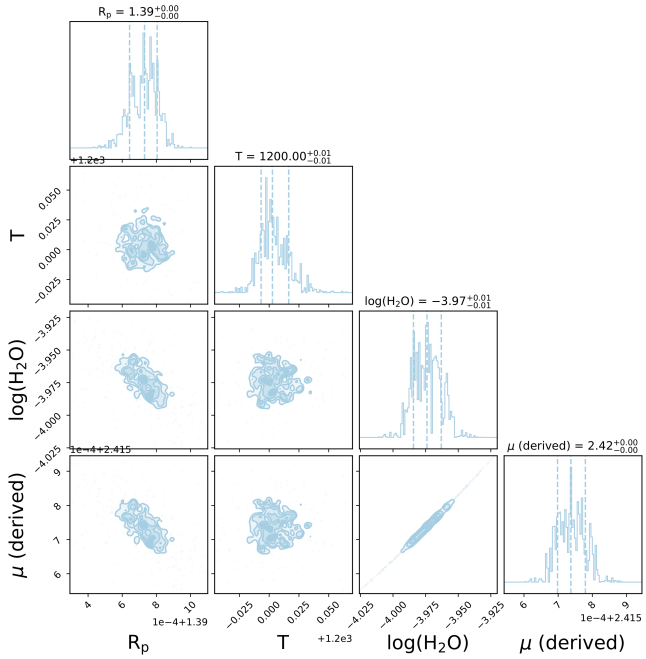

Fig. 4. Posterior distributions of the retrieved parameters from the synthetic transmission spectrum: temperature (T), planetary radius (R_p), and water abundance ($\log_{10}(H_2O)$).

5. Conclusion

In this exoplanetary study, we focused on two key aspects: learning how to detect transiting exoplanets using machine learning, and studying the atmosphere of a specific exoplanet, HD 209458b using atmospheric modelling and retrieval techniques.

In the first part, we used machine learning models, including SVM and ANN, to study how exoplanet transits can be detected in stellar light curves. These models were trained and tested on both synthetic transits and real observational data obtained from Kepler. We observed that ANN performed slightly better than the SVM overall. In both the cases, the precision of the training data was noticeably higher than on the validation (test) data. Additionally, both models performed well on the synthetic data but struggled with the real dataset, highlighting signs of over-fitting and the difficulty of working with real observational data. This part of the study helped us understand the challenges in transit detection and how machine learning can be a useful tool though limited.

In the second part, we focused specifically on HD 209458b, a well-known hot Jupiter with rich observational data. We used TauREx 3 to model and retrieve its atmospheric properties. First, we created a synthetic transmission spectrum using known planetary parameters, assuming a clear atmosphere. The retrieval on this synthetic spectrum successfully recovered accurate values for temperature, radius, and water abundance, confirming the accuracy our model. We then applied the same retrieval method to real observational data of HD 209458b. The results showed a lower atmospheric temperature, consistent planetary

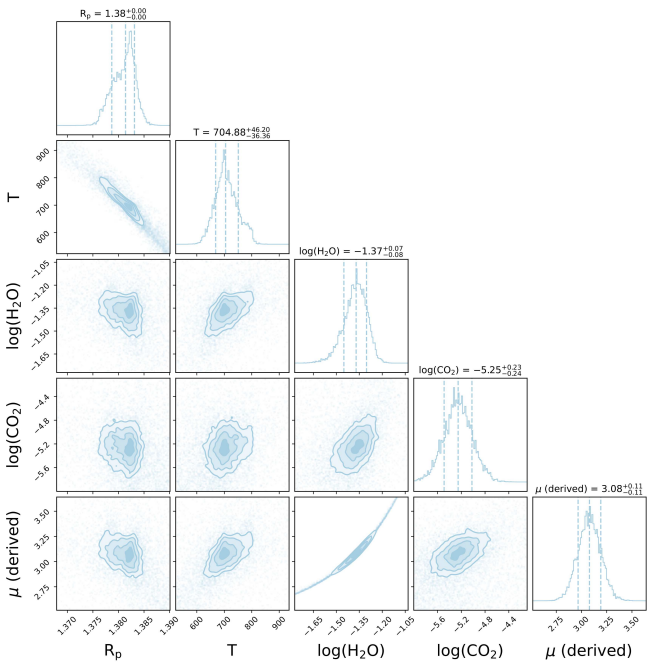

Fig. 5. Posterior distributions of the retrieved parameters from the observed transmission spectrum: temperature (T), planetary radius (R_p), water abundance ($\log_{10}(H_2O)$) and carbon dioxide ($\log_{10}(CO_2)$).

radius, and detectable amounts of H_2O and CO_2, matching findings from previous studies.

Overall, this paper combined a study of detection methods using machine learning with a detailed atmospheric analysis of HD 209458b. While machine learning offers valuable insights into transit detection techniques, tools like TauREx are essential for understanding exoplanet atmospheres in detail. Together, these approaches provide a strong foundation for future exoplanet characterization efforts.

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